

Math261 Midterm Exam 1

Exam 1 covers the following topics, and the corresponding practice problems are listed.

1. Dot product/cross product
 - Computation (Practice Q1, 3; HW1: 1-4)
 - Properties (Q5, 6; HW1: #6, 9)
 - Determine perpendicular or parallel vectors (Q4)
2. Parametric equations for lines
 - Finding parametric equations (Q7, Q8; HW3: 10-13)
 - Lines Intersection (Q9; HW3: 14-15)
3. Parametric equations of curves
 - Line segments (Q19; HW1 Q18)
 - Circles (Q15, 16; HW1: 16)
 - Ellipses (Q17,18, HW1: 17)
4. Tangent vectors to parametric curves (Q20; HW1: 20-22)
5. Equation of planes
 - Finding equation of a plane given three points on the plane, or one point and a normal vector; (Q10,11,12; HW2: #2-6)
 - Finding parametric equations for the line of intersection of two planes (Q13,14; HW2:11-12)
 - Finding the intersection point(s) of a line and a plane (Q7, Q8; HW2: 7-9)
6. Partial derivative
 - Computing first-order, and second-order partial derivatives (Q21-25; HW2:13-15)
 - Chain Rule (Q41-46; HW2: 16-21)
 - Determine the sign of a partial derivative by visualizing a surface graph (Q48-49)
7. Directional derivative and gradient vector
 - Computing the gradient vector (HW3: #2-3); finding the directions of steepest increase or decrease (Q29, 30, 31; HW3: #4-8); draw the gradient vector relative to level curves (perpendicular to the curve, pointing toward increasing values) (Q49-50a,b)
 - Computing directional derivative in a given direction (Q26; HW3: #9-10, 13-15); Using its sign to determine whether the function is increasing or decreasing (Q27, 28; HW3: 11-12), and confirm that conclusion against the graph (Q49-50c)
 - Finding the equation of the tangent plane to a surface (Q32-35; HW3: 22-25)
8. Critical points and local extreme values
 - Finding critical points, and classify the type for each point (Q36-40; HW3: 16-25)